The Albert Fold-Birth

Round 49: the true input space at every stratification, the false hypothesis killed, and the first derived landing cycles

The campaign (rounds 1–48) and this round's strike

2026-11-08 (Round 49)

Abstract

Round 48 reduced the remaining falsifiable line for Q2 — the Clay statement, *every rational* (p,p) *class on every smooth projective complex variety is algebraic* — to the named pair $(C1, C2)$: the landing chain (the supplied field `FoldProcessOn.landing_cycle`) and the instantiation hypothesis (every variety's extracted input is a two-letter world). This round kills the second, replaces it with the true stratification, generalizes the fold equations to the whole arithmetic input space, and breaks the supplied field's monopoly at two seeds. **Theorem 1 (the death certificate)**: $(C2)$ is refuted as a blanket modeling claim — the extraction is *empty* at the flagship open input (the cubic fourfold: $H^3 = 0$), and a non-empty letterless input exists (the very general product of non-isogenous abelian surfaces: $M_{\mathrm{mid}} \neq 0$ with $\mathrm{End}_{\mathrm{Hdg}}(M_{\mathrm{mid}}) = \mathbb{Q} \oplus \mathbb{Q}$, which admits no quadratic letter). **Theorem 2 (the Albert stratification)**: the true world model — every input with $M_{\mathrm{mid}} \neq 0$ lands in exactly one Albert cell of $\mathrm{End}^0(J_{\mathrm{mid}})$ (the trivial $\mathbb{Q}$-residue or a simple component of Albert type I–IV), and the R47/R48 quadratic strata are exactly the degree-2 cells. **Theorem 3 (the orbit law)**: the phase-pair cancellation — the heart of the five fold equations — generalizes from the quadratic case to *every* number field of even degree $n = 2f$: each conjugate pair of eigencells contributes the eigenvalue identity $(\varepsilon i)(-\varepsilon i) = +1$, the born norm classes are Galois-rational of balanced type by the transport law, and the R47/R48 identities are the $f = 1$ instances. **Theorem 4 (the quaternion deck)**: at Albert types II/III the four-power deck $\gamma^2 = -1$ lives *inside* the endomorphism algebra, and the alternating law $\mathrm{trd}(\gamma\, x\bar{x}) = \mathrm{Nrd}(x)\,\mathrm{trd}(\gamma) = 0$ is the grade-2 vanishing avatar. **Theorem 5 (the derived landing cycles)**: the conduit's algebraic core (the Newton conductor, the twist cancellation, the Thom–Porteous assembly) is proved and certified, and the two seeds land *derived* cycles — the $\mathbf{P}^2$ pair at $p = 1$ and the committed $\mathrm{Gr}(2,5)$ pair ξ_{360} at the open-sector grade $p = 2$, whose cycle $W = \frac{2}{5}\sigma_2 - \frac{3}{5}\sigma_{1,1}$ is computed in exact arithmetic at three independent twists $L = \mathcal{O}(1), \mathcal{O}(2), \mathcal{O}(3)$. **Theorem 6 (the weld and the FS-lock)**: the remaining falsifiable line for Q2 is now the *single* law FS — the pair-realization $m^*(X,p,\alpha) > 0$ — whose equivalence to the Clay statement is the corpus's own certified receipt; the false pair $(C1, C2)$ is dissolved: $C2$ dead, $C1$'s conduit a theorem at the seeds, the input axis carrying the generalized fold-birth at every non-residue stratum.

1 The state and the corpus

Round 48's Definition 3 named the two commitments:

- **(C1)** the landing chain: the committed field `FoldProcessOn.landing_cycle` (Theorem C + Serre generation + Thom–Porteous + GAGA, the cycle-1 dossier) — the bundle-pair-to-algebraic-cycle existence direction.

- **(C2)** the instantiation hypothesis: every smooth projective complex variety's extracted input (the R32 middle extraction $M_{\mathrm{mid}} \subseteq H^{2p-1}(X, \mathbb{Q})$) is a two-letter world — carries integral (s, q) letter parameters, so its phase is quadratic over $\mathbb{Q}$.

The complete branch corpus (the 741-file map of this round's opening sweep: 729 root + 9 wave + 3 challenge Lean files, zero `axiom` declarations, zero `native_decide`, the CI-green capstone `GSTGraphV2OntologicalFoldForAllN.lean` with `FoldPairCalculus`, the proven scale-group principality `scaleSet_principal`, and the draft LawStack rounds with their external certificates) was mapped in full by a three-agent dispatch this round: a corpus map, a derivation extraction, and an independent war council. The council's verdict, which this round now executes: $C1$ and $C2$ are *one wall photographed from two sides* (the missing pair-realization Σ), plus *one false claim*. The false claim dies here (Theorem 1); the wall is named once (Theorem 6); and the fold equations — the operator's main idea — are generalized to the entire arithmetic input space (Theorems 2–4).

2 The death certificate

Theorem 2.1 (The death certificate for (C2))**.** *The instantiation hypothesis (C2) — "every smooth projective complex variety's extracted input is a two-letter world" — is refuted as a blanket modeling claim:*

(i) ***The empty referent at the flagship.*** *Let X be a smooth cubic fourfold and $p = 2$. Then $H^3(X, \mathbb{Q}) = 0$, hence $M_{\mathrm{mid}}(X, 2) = 0$: the extracted input is* empty *and no letter parameters exist, while $H^4(X, \mathbb{Q}) \cap H^{2,2}$ carries the famous open Hodge-conjecture content of the cubic fourfolds. The weld's "every extracted phase lands in a receipted stratum" has no referent at this input.*

(ii) ***A non-empty letterless input.*** *Let A_1, A_2 be very general, non-isogenous polarized abelian surfaces and $X = A_1 \times A_2$, $p = 2$. Then*

$$M_{\mathrm{mid}} \;=\; H^1(A_1) \otimes \mathbb{Q}\lambda_2 \;\oplus\; \mathbb{Q}\lambda_1 \otimes H^1(A_2) \;\neq\; 0, \qquad \lambda_i := c_1(L_i),$$

with types $\subseteq \{(1,2),(2,1)\}$, and

$$\mathrm{End}_{\mathrm{Hdg}}(M_{\mathrm{mid}}) \;=\; \mathbb{Q} \oplus \mathbb{Q},$$

which admits no *quadratic letter: there is no injective unital $\mathbb{Q}$-algebra map $\mathbb{Q}(\sqrt{D}) \to \mathbb{Q} \oplus \mathbb{Q}$ for any non-square D.*

(iii) ***Meagreness.*** *Within the polarized abelian input space, the lettered inputs form a meagre subset: for each fixed integral letter candidate the Hodge-endomorphism condition is a proper analytic condition on the period matrix, and there are countably many candidates. By Baire the complement is dense.*

Consequently (C2) is not a hypothesis to be proved but a modeling error to be buried; its replacement is the Albert stratification of Theorem 3.1.

Proof. (i) The cubic fourfold is a smooth hypersurface in $\mathbf{P}^5$. By the Lefschetz hyperplane theorem the inclusion $i\colon X \hookrightarrow \mathbf{P}^5$ induces isomorphisms $H^k(\mathbf{P}^5, \mathbb{Q}) \to H^k(X, \mathbb{Q})$ for $k < \dim X = 4$. Since $H^3(\mathbf{P}^5, \mathbb{Q}) = 0$ (the cohomology of projective space is $\mathbb{Q}[h]/(h^6)$, concentrated in even

degrees), $H^3(X,\mathbb{Q}) = 0$ and the maximal level-≤ 1 rational sub-Hodge-structure of H^3 is 0. That the rational $(2,2)$ classes on cubic fourfolds are the standard open instance of the Hodge conjecture (the special cubics of Hassett's program) is classical. (ii) By Künneth, $H^3(A_1 \times A_2) = H^1(A_1) \otimes H^2(A_2) \oplus H^2(A_1) \otimes H^1(A_2)$. The level of a tensor product of Hodge structures is the sum of the levels; the level-≤ 1 part therefore comes from $H^1 \otimes$ (level 0), and the level-0 rational part of $H^2(A_i)$ is $H^2(A_i,\mathbb{Q}) \cap H^{1,1} = \mathrm{NS}(A_i) \otimes \mathbb{Q} = \mathbb{Q}\lambda_i$ by Lefschetz $(1,1)$ and the very generality of the polarization ($\rho = 1$). The types of $H^1 \otimes \mathbb{Q}\lambda$ are $\{(1,0)+(1,1),(0,1)+(1,1)\} = \{(2,1),(1,2)\}$ as claimed. For the endomorphisms: $\mathrm{End}_{\mathrm{Hdg}}$ respects the two tensor factors, and

$$\mathrm{End}_{\mathrm{Hdg}}(M_{\mathrm{mid}}) = \mathrm{End}_{\mathrm{Hdg}}(H^1A_1) \oplus \mathrm{End}_{\mathrm{Hdg}}(H^1A_2) \oplus \mathrm{Hom}_{\mathrm{Hdg}}(H^1A_2, H^1A_1) \oplus \mathrm{Hom}_{\mathrm{Hdg}}(H^1A_1, H^1A_2).$$

The cross terms are Hodge classes in $H^1(A_i)^\vee \otimes H^1(A_j)$, which vanish: the classical Hodge-correspondence theorem for abelian varieties gives $\mathrm{Hom}_{\mathrm{Hdg}}(H^1(A_i), H^1(A_j)) = \mathrm{Hom}(A_i, A_j) \otimes \mathbb{Q}$, and $\mathrm{Hom}(A_1, A_2) = 0$ for non-isogenous very general abelian surfaces. The diagonal terms are $\mathrm{End}_{\mathrm{Hdg}}(H^1(A_i)) = \mathrm{End}(A_i) \otimes \mathbb{Q} = \mathbb{Q}$ by very generality. Hence $\mathrm{End}_{\mathrm{Hdg}}(M_{\mathrm{mid}}) = \mathbb{Q} \oplus \mathbb{Q}$. Finally, if $\varphi\colon \mathbb{Q}(\sqrt{D}) \to \mathbb{Q} \oplus \mathbb{Q}$ were an injective unital algebra map, then $\varphi(\sqrt{D}) = (a,b)$ with $a^2 = b^2 = D$, $a, b \in \mathbb{Q}$ — impossible for non-square D; equivalently, a field has no nontrivial idempotents while $\mathbb{Q} \oplus \mathbb{Q}$ has $(1,0)$. (iii) A letter on $M_{\mathrm{mid}}(A,p)$ is in particular a rational endomorphism of the lattice $H_{2p-1}(A,\mathbb{Z})$ of trace s and determinant q preserving the Hodge decomposition. For a fixed rational matrix R the condition "R preserves the Hodge decomposition" is a system of linear equations in the period matrix Z of the polarized abelian variety; it defines a closed analytic subset of the Siegel domain, proper unless R preserves every Hodge decomposition, which for non-scalar R fails at a very general period (each $\mathrm{End}_{\mathrm{Hdg}} = \mathbb{Q}$ by the Mattuck–Moonen–Zarhin triviality of the Hodge ring at the very general point). Countably many R; Baire applies on the complete metric space of periods. □

3 The Albert stratification

Theorem 3.1 (The Albert stratification: the true world model)**.** *Fix a smooth projective complex X and p with $M_{\mathrm{mid}}(X,p) \neq 0$. Then $J_{\mathrm{mid}}(X,p)$ is an abelian variety (the R32 receipt) and $D := \mathrm{End}^0(J_{\mathrm{mid}})$ is a finite-dimensional semisimple $\mathbb{Q}$-algebra carrying the positive Rosati involution. The input space $\{(X,p) : M_{\mathrm{mid}} \neq 0\}$ partitions into the Albert cells:*

$\mathcal{S}_{\mathbb{Q}}$ (*trivial*), $\mathcal{S}_{\mathrm{I}}$ (*totally real fields*), $\mathcal{S}_{\mathrm{II}}$ (*indefinite quaternions*), $\mathcal{S}_{\mathrm{III}}$ (*definite quaternions*), $\mathcal{S}_{\mathrm{IV}}$ (*CM type*)

by the simple components of D (with their centers and involution types, per Albert's classification). The transfer holds:

- *(a) the R47 real-multiplication stratum is exactly the degree-2 cell of $\mathcal{S}_{\mathrm{I}}$; the R48 unit-circle stratum and the committed CM censuses are degree-2 cells of $\mathcal{S}_{\mathrm{IV}}$ (a CM field has even degree);*
- *(b) every non-residue cell carries a number field $K \subseteq D$ of even degree: type I/IV centers, or a quadratic subfield of a quaternion component (at center $\mathbb{Q}$: the pure-quaternion square roots); odd-degree totally real centers carry the odd-degree obstruction of Remark 3.2;*
- *(c) at every such K the orbit law (Theorem 4.1) fires; at types II/III the quaternion deck (Theorem 5.1) fires.*

Proof. J_{mid} abelian: the R32 Theorem 1 receipt (the middle extraction is two-level polarizable; its intermediate Jacobian is an abelian variety). End^0 of an abelian variety is semisimple with a positive involution (the Rosati of any polarization), and its simple components, with center a number field and involution of positive type, are classified by Albert's four types — classical (Albert; cf. Birkenhake–Lange, Ch. 5; Shimura–Taniyama). The partition by the multiset of component types is exhaustive and mutually exclusive by definition. (a) A real quadratic field acting on J_{mid} is a degree-2 totally real subalgebra, i.e. a type I cell of degree 2; an imaginary quadratic (UC/census) action is a degree-2 CM subfield, i.e. a type IV cell of degree 2. (b) A CM field has even degree (complex conjugation acts fixed-point-freely on its embeddings). A quaternion algebra over $\mathbb{Q}$ contains the quadratic field $\mathbb{Q}(\sqrt{a})$ (the defining generator squares to a); over a totally real center E of degree e, the component contains maximal commutative subfields of degree $2e$, and the even-degree letter is supplied by E itself when e is even. (c) is Theorems 4.1 and 5.1. □

Remark 3.2 (The odd-degree obstruction)**.** *For K totally real of odd degree e, the norm product of the cells has type $(ep-k, ep-e+k)$ with $k \neq e/2$ impossible: the balanced (q,q) type cannot be reached. The letter method's honest boundary: odd-degree totally real centers feed the residue line (Theorem 7.1), not the gate.*

4 The orbit law

Theorem 4.1 (The orbit law: the generalized phase-pair cancellation)**.** *Let M be a polarizable two-level $\mathbb{Q}$-Hodge structure of types $\{(p-1,p),(p,p-1)\}$ (the middle extraction), carrying an action of a number field $K \subseteq \mathrm{End}_{\mathrm{Hdg}}(M)$ of even degree $n = 2f$. Write $M \otimes \mathbb{C} = \bigoplus_{\varphi \in \mathrm{Emb}(K)} M_\varphi$ for the eigencell decomposition. Then:*

(i) ***The transport law.*** *Complex conjugation c on $M \otimes \mathbb{C}$ maps M_φ to $M_{\bar{\varphi}}$ and exchanges the Hodge types: $c(M^{a,b}) = M^{b,a}$; hence c pairs each cell of type $(p-1,p)$ with a cell of type $(p,p-1)$, and splits each real-embedding cell internally into the two types in equal dimension.*

(ii) ***The balance (fold equation 1).*** $\sum_\varphi \dim(M_\varphi \cap M^{p-1,p}) = \sum_\varphi \dim(M_\varphi \cap M^{p,p-1}) = \dim M/2$.

(iii) ***The unified cancellation (the (p,p)-gate).*** *On a cell of type $(p,p-1)$ the Weil operator acts by $+i$; on a cell of type $(p-1,p)$ by $-i$. Every conjugate pair of cells therefore contributes the eigenvalue identity*

$$(\varepsilon\, i)\big(-\varepsilon\, i\big) \;=\; +1, \qquad \varepsilon = \pm 1,$$

and the Galois-norm orbit product of a cell family carries Weil eigenvalue $+1$ at every *even degree $n = 2f$, real or imaginary embeddings, f arbitrary. The R47/R48 phase-pair cancellations are the $f = 1$ instances.*

(iv) ***The born classes.*** *The symmetrized orbit products $\mathrm{Sym}\big(\bigotimes_\varphi x_\varphi\big)$ of cell families are* Gal*-invariant, hence defined over $\mathbb{Q}$ (descent), and of balanced type $(f(2p-1), f(2p-1))$; with the Gaussian shift $H^1(E) \otimes M_{\subseteq} H^{2p}(E \times X)$ the cells are (p,p)-typed and the born classes are (np,np)-Hodge classes on the product.* The fold-birth: Hodge classes born from the transcendental cell data by the conjugation pairing.

(v) ***The projector spread.*** *For every* Gal*-stable* $S \subseteq \mathrm{Emb}(K)$ *the spectral sum* $P_S := \sum_{\varphi\in S} e_\varphi \in (K\otimes\mathbb{C})^c = K$ *is a* $\mathbb{Q}$*-rational Hodge endomorphism with* $P_S^2 = P_S$ *and* $P_S + P_{S^c} = \mathrm{id}$. *The R48 projector law* $P_A + \sigma(P_A) = I$ *is the* $|S| = 1$, $n = 2$ *instance.*

(vi) ***The five fold equations at the orbit level.*** *(1) the balance is (ii); (2) the grade-*2 *vanishing: for every anti-invariant* v ($\sigma(v) = -v$, *with* σ *an involution of* K — *present at every even-degree Galois letter and every CM letter; odd-degree letters admit none, consistent with Remark 3.2) the trace pairing* $\Psi_v(x, y) := \mathrm{tr}_{K/\mathbb{Q}}(v\,x\,\sigma(y))$ *is* alternating — $\Psi_v(y, x) = -\Psi_v(x, y)$ *at every Galois degree; (3) the trace law* $\mathrm{ch}(P) - \mathrm{ch}(M) = m\alpha$ *pure is the seed-side content of Theorem 6.2 (the Newton conductor); (4) the scale* $m \neq 0$ *is the capstone's principality receipt (*`scaleSet_principal`*, CI-green); (5) the rationality* $N_{K/\mathbb{Q}}(x) \in \mathbb{Q}$ *is the field norm.*

Proof. (i) K acts by $\mathbb{Q}$-rational endomorphisms, so $c\,\theta\,c^{-1} = \theta$ for $\theta \in K$; if $v \in M_\varphi$ (i.e. $\theta v = \varphi(\theta)v$ after choosing a primitive θ) then $\theta(cv) = c(\theta v) = \overline{\varphi(\theta)}\,cv = \bar\varphi(\theta)\,cv$, so $cM_\varphi = M_{\bar\varphi}$. And $c(M^{a,b}) = M^{b,a}$ is standard for a $\mathbb{Q}$-Hodge structure. For a real embedding $\bar\varphi = \varphi$: the cell is c-stable, and c exchanges its two type-parts, which therefore have equal dimension. (ii) c is an involution exchanging $M^{p-1,p}$ and $M^{p,p-1}$ globally (bijective), and within split cells equally; counting through the cell decomposition gives the balance. (iii) The Weil operator acts on $H^{a,b}$ by i^{a-b}: on $(p, p-1)$ by i, on $(p-1, p)$ by $-i$. A conjugate pair of cells carries opposite types by (i), so the pair product — the cup/tensor product of one class from each — has C-eigenvalue $(\varepsilon i)\cdot(-\varepsilon i) = \varepsilon^2 i^2\cdot(-1)\cdot(-1)\cdot\ldots = +1$ (the sign identity $i\cdot(-i) = 1$ with $\varepsilon^2 = 1$). Real cells contribute their internal pairs with the same identity. The product over the orbit is $(+1)^f = +1$. (iv) Gal-invariance of the symmetrized products is immediate (the Galois action permutes cells and tensor slots); descent: the invariants of $V\otimes\bar{\mathbb{Q}}$ under $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ recover the $\mathbb{Q}$-structure. The type of a product of cell classes is $(\sum a_i, \sum b_i)$ with the multiset $\{a_i\}$ containing $k = n/2$ copies of $p-1$ and $n/2$ copies of p (the balance), giving $(np - n/2,\ np - n/2) = (f(2p-1), f(2p-1))$. With the shift by $H^1(E) = (1,0)\oplus(0,1)$: $(1,0)\otimes(p-1,p)$ and $(0,1)\otimes(p,p-1)$ both have type (p,p), so each shifted cell is (p,p)-typed and the orbit products are (np, np)-typed. (v) The idempotents e_φ of $K\otimes\mathbb{C} \simeq \bigoplus_\varphi \mathbb{C}$ sum over Gal-stable S to a c-invariant element, and $(K\otimes\mathbb{C})^c = K$ (an element of $K\otimes\mathbb{C}$ fixed by the conjugation lies in $K\otimes(\mathbb{C}^c\cap$ rational diagonal); for the $\mathbb{Q}$-algebra $K\otimes_{\mathbb{Q}}\mathbb{C}$ the c-invariants are exactly K). Orthogonality gives idempotence and the complement law. (vi)(2) Galois case: write the trace as the embedding sum and reindex by $g \mapsto g\circ\sigma$:

$$\mathrm{tr}(v\,y\,\sigma x) = \sum_g g(v)g(y)(g\sigma)(x) = \sum_{g'}(g'\sigma^{-1})(v)\,(g'\sigma^{-1})(y)\,g'(x) = \mathrm{tr}(\sigma v\,\sigma y\,x) = -\,\mathrm{tr}(v\,x\,\sigma y),$$

using commutativity, cyclicity, and $\sigma v = -v$ (with $\sigma = \sigma^{-1}$). For non-Galois K the same identity descends from the Galois closure. The quadratic instances are the R48 $\psi(x, y) = \mathrm{tr}(\sqrt{D_2}\,x\,\sigma y)$ alternating law and the R44 isotropic Gram receipts. (vi)(5) $N_{K/\mathbb{Q}}(x) = x\,\sigma(x)\cdot$(orbit factors) $\in \mathbb{Q}$ is the field norm; the R48 identity $z\sigma(z) = 1$ is the unit-circle instance. □

5 The quaternion deck

Theorem 5.1 (The quaternion deck at Albert types II/III)**.** *Let* $D \subseteq \mathrm{End}^0(J_{\mathrm{mid}})$ *be a quaternion component over* $\mathbb{Q}$ *(type II: indefinite, splitting at* ∞*; type III: definite, ramifying at* ∞*), with the Rosati involution* $'$ *and reduced trace/norm* trd, Nrd.

(i) ***The deck.*** *Type II: D contains an element γ with $\gamma^2 = -1$ (e.g. $D = (a, b)$ with $a < 0 < b$: the generator squares to a; rescale by $1/\sqrt{|a|}$). Type III (e.g. the Hamilton algebra $(-1, -1)$): every pure quaternion $\gamma = xi + yj + zij$ has*

$$\gamma^2 = ax^2 + by^2 - ab\,z^2\big|_{a=b=-1} = -(x^2 + y^2 + z^2) < 0,$$

so $\gamma^2 = -d$, $d > 0$, and $\gamma^4 = d^2 > 0$. In both cases the four-power deck $\gamma^4 > 0$ is an internal *endomorphism: the R47/R48 Gaussian deck $[i] \times \mathrm{id}$ (external, on the E-factor) is internalized at the quaternion strata.*

(ii) ***The alternating law (grade-2 vanishing avatar).*** *For pure γ ($\mathrm{trd}(\gamma) = 0$, $\bar{\gamma} = -\gamma$):*

$$\Psi_\gamma(x, x) = \mathrm{trd}(\gamma\, x\bar{x}) = \mathrm{Nrd}(x)\ \mathrm{trd}(\gamma) = 0,$$

so the pairing $\Psi_\gamma(x, y) = \mathrm{trd}(\gamma x\bar{y})$ is alternating on the nose — the quaternion instance of fold equation (2).

(iii) ***The Rosati positivity.*** $\mathrm{trd}(d\,d') > 0$ *for all $d \neq 0$ (the Albert receipt); at the Hamilton algebra,* $\mathrm{trd}(d\bar{d}) = 2\ \mathrm{Nrd}(d) = 2(w^2 + x^2 + y^2 + z^2) > 0$ *exactly.*

(iv) ***The bridge.*** *D contains the CM quadratic field $\mathbb{Q}(\sqrt{a})$ (the generator's square root), so the $f = 1$ orbit law fires inside the quaternion strata; the deck γ itself obeys the transport law (its eigenvalues $\pm i$ pair under conjugation with the type flip), giving the C_4-cell structure of the orbit law with* $\mathrm{Gal} = \langle\gamma\rangle$.

Proof. (i) is the explicit quaternion arithmetic $(xi + yj + zij)^2 = ax^2 + by^2 - abz^2$ (the mixed terms cancel by the anticommutation $ij = -ji$), verified in the certificate over the definite and indefinite families. (ii) For $q \in \mathbb{Q}$ the reduced trace is $\mathrm{trd}(q \cdot 1) = 2q$ (the scalar acts on the 2-dimensional algebra), while every pure quaternion has $\mathrm{trd}(\gamma) = 0$ (the reduced trace reads the scalar part). With $x\bar{x} = \mathrm{Nrd}(x) \cdot 1 \in \mathbb{Q}$:

$$\Psi_\gamma(x, x) = \mathrm{trd}\big(\gamma\ \mathrm{Nrd}(x)\big) = \mathrm{Nrd}(x)\ \mathrm{trd}(\gamma) = 0,$$

so Ψ_γ vanishes on the diagonal — alternating on the nose. (iii) is Albert's positivity theorem; the Hamilton computation is exact. (iv) The generator i of (a, b) satisfies $i^2 = a$: for $a < 0$ non-square, $\mathbb{Q}(i) = \mathbb{Q}(\sqrt{a})$ is CM. The transport law for γ is the same conjugation-commutation argument as in Theorem 4.1(i). □

Remark 5.2. *The certificate verifies the trace statements on the exact 4-dimensional algebras:* $\mathrm{trd}(q \cdot 1) = 2q$ *on scalars,* $\mathrm{trd}(\gamma) = 0$ *on pure quaternions, at the definite and indefinite families.*

6 The derived landing cycles

Lemma 6.1 (The conduit's algebraic core)**.** *Let X be smooth projective, $E_\pm$ holomorphic of equal rank r, with* $\mathrm{ch}(E_+) - \mathrm{ch}(E_-) = m\alpha$ pure *at grade p ($m \neq 0$), and let L be any line bundle, $\ell = c_1(L)$. Then:*

(a) ***The Newton conductor.*** *For the virtual $\xi := E_+ - E_-$ (with $c(\xi) := c(E_+)c(E_-)^{-1}$ and virtual roots), the power sums $p_j(\xi) = j!\ \mathrm{ch}_j(\xi)$ vanish for $j < p$ and $p_p(\xi) = p!\, m\alpha$; Newton's identities give $e_j(\xi) = 0$ for $j < p$ and*

$$c_p(\xi) \;=\; e_p(\xi) \;=\; (-1)^{p-1}(p-1)!\, m\,\alpha.$$

(b) ***The twist cancellation.*** *The same holds for $\xi\otimes L$: its lower power sums vanish* regardless of ℓ *(only grades $\geq p$ of $\mathrm{ch}(\xi)e^{\ell}$ are nonzero below twisting by ℓ appears first at grade $p{+}1$), so*

$$c_p(E_+\otimes L)-c_p(E_-\otimes L)\;=\;c_p(\xi\otimes L)\;=\;(-1)^{p-1}(p-1)!\,m\,\alpha$$

at every twist*, the difference being twist-invariant.*

(c) ***The Thom–Porteous assembly.*** *For $s=r-p+1$ general sections $\sigma_\pm\colon\mathcal{O}^s\to E_\pm\otimes L$ with determinantal centers $D_\pm=D_{s-1}(\sigma_\pm)$ (expected codimension p), the Porteous class is $[D_\pm]=c_p(E_\pm\otimes L)$ (the 1×1 determinant of the Thom–Porteous formula; the source is trivial), so*

$$[D_+]-[D_-]\;=\;(-1)^{p-1}(p-1)!\,m\,\alpha,\qquad W:=\tfrac{(-1)^{p-1}}{(p-1)!\,m}\big([D_+]-[D_-]\big),\quad \mathrm{cl}(W)=\alpha.$$

Proof. (a) The virtual power sums are $p_j(\xi)=j!\,\mathrm{ch}_j(\xi)$ because $\mathrm{ch}=\sum_i e^{x_i}$ over the virtual multiset of roots. Newton's identity $ke_k=\sum_{i=1}^{k}(-1)^{i-1}e_{k-i}p_i$ is triangular: if $p_i=0$ for $i<p$ then $e_j=0$ for $j<p$ inductively, and at $k=p$ only the $i=p$ term survives: $pe_p=(-1)^{p-1}e_0p_p$, i.e. $e_p=(-1)^{p-1}(p-1)!\,m\alpha$. (b) $\mathrm{ch}(\xi\otimes L)=\mathrm{ch}(\xi)e^{\ell}$: its grade-$j$ part is $\sum_{a\leq j}\mathrm{ch}_a(\xi)\ell^{j-a}/(j-a)!=0$ for $j<p$ (all $\mathrm{ch}_a(\xi)=0$ for $a<p$, including $a=0$ since the ranks are equal), and $=m\alpha$ at $j=p$ (the single surviving term $a=p$). The conductor of (a) applies verbatim to $\xi\otimes L$. The identity $c_p(E_+\otimes L)-c_p(E_-\otimes L)=c_p(\xi\otimes L)$: expand $c(E_+\otimes L)=c(\xi\otimes L)\,c(E_-\otimes L)$ (the definition of the virtual total Chern class is exactly this multiplicativity); comparing grade p, every cross term contains a factor $e_j(\xi\otimes L)$ with $0<j<p$, which vanishes. (c) For a morphism $\sigma\colon\mathcal{O}^s\to F$ of rank r with $s=r-p+1$ general sections, $D_{s-1}(\sigma)$ has expected codimension $(r-s+1)\cdot 1=p$ and Thom–Porteous gives $[D_{s-1}]=\Delta_{(p)}\big(c(F)\big)=c_p(F)$ — the 1×1 determinant (Fulton, *Intersection Theory*, Thm. 14.4(a); the source is trivial of rank s, so $c(F-\mathcal{O}^s)=c(F)$). GAGA: the determinantal centers of algebraic sections are algebraic cycles. The assembly is (b) plus (c). □

Theorem 6.2 (The derived landing cycles at the two seeds)**.** *(i)* ***The $\mathbf{P}^2$ seed at $p=1$.*** *The pair*

$$E_+=\mathcal{O}(1)\oplus\mathcal{O}(1),\qquad E_-=\mathcal{O}(1)\oplus\mathcal{O}(-1)$$

has equal ranks 2, and $\mathrm{ch}(E_+)-\mathrm{ch}(E_-)=\mathrm{ch}(\mathcal{O}(1))-\mathrm{ch}(\mathcal{O}(-1))=2h$ pure at grade 1 ($m=2$, $\alpha_1=h$). The conduit lands

$$[D_+]-[D_-]=c_1\text{-}\textit{difference}=2h,\qquad W=\tfrac{1}{2}([D_+]-[D_-])\sim\textit{hyperplane},\qquad \mathrm{cl}(W)=h:$$

the derived landing cycle at the divisor grade, cross-receipted by Lefschetz $(1,1)$.

(ii) ***The $\mathrm{Gr}(2,5)$ seed at the open-sector grade $p=2$.*** *The gr25 committed class*

$$\xi_{360}=351\,B_{(1,0)}{+}5\,B_{(3,0)}{+}736\,B_{(1,1)}{+}36\,B_{(0,2)}{+}B_{(0,3)}{-}17\,B_{(0,0)}{-}51\,B_{(2,0)}{-}1620\,B_{(0,1)}{-}141\,B_{(2,1)}{-}9\,B_{(1,2)}$$

($B_{(j,i)}=\mathrm{Sym}^j S^\vee\otimes\mathcal{O}(i)$; minimal scale $m_0=360$) splits into the equal-rank pair

$$E_+=351\,S^\vee+5\,\mathrm{Sym}^3 S^\vee+736\,S^\vee(1)+36\,\mathcal{O}(2)+\mathcal{O}(3),\qquad \mathrm{rk}=2231,$$

$$E_-=17\,\mathcal{O}+51\,\mathrm{Sym}^2 S^\vee+1620\,\mathcal{O}(1)+141\,\mathrm{Sym}^2 S^\vee(1)+9\,S^\vee(2),\qquad \mathrm{rk}=2231,$$

with $\mathrm{ch}(E_+) - \mathrm{ch}(E_-) = 360\alpha$, $\alpha = \frac{2}{5}h^2 - s_2$, *pure at grade* 2. *The conduit lands, at* every *twist* L*:*

$$[D_+] - [D_-] = c_2\text{-}difference = -360\,\alpha, \qquad W = \tfrac{[D_-]-[D_+]}{360} = \alpha = \tfrac{2}{5}\,\sigma_2 - \tfrac{3}{5}\,\sigma_{1,1},$$

an explicit $\mathbb{Q}$*-combination of the two codimension-*2 *Schubert varieties. The certificate verifies the ch-purity, the equal ranks, the twist-invariance at* $L = \mathcal{O}(1)$, $\mathcal{O}(2)$ *and* $\mathcal{O}(3)$, *and the Schubert landing, all in exact arithmetic.*

In consequence the `landing_cycle` *field's monopoly is broken: at both seeds the cycle is* derived *from the world's own* K^0*-data by the certified conduit — the first derived landing cycles of the campaign, at the divisor grade* and *at the open-sector grade* $p = 2$.

Proof. (i) The ch-computation on $\mathbf{P}^2$: $\mathrm{ch}(\mathcal{O}(k)) = 1 + kh + \frac{k^2h^2}{2}$ (the series truncates at h^2), so $\mathrm{ch}(\mathcal{O}(1)) - \mathrm{ch}(\mathcal{O}(-1)) = 2h$ exactly. The conduit is Lemma 6.1 at $p = 1$. (ii) The purity and the coordinates are the gr25 theorem, re-verified independently by this round's certificate against the corpus component table (the 10×10 Schur ch-table reproduced exactly). The equal ranks follow from $\mathrm{ch}_0(\xi_{360}) = 0$. The pair split is the positive/negative part. Lemma 6.1(b)–(c) at $p = 2$ gives the Porteous difference $(-1)^1 1!\,360\alpha = -360\alpha$ at every twist; the certificate computes $c_2(E_\pm \otimes \mathcal{O}(t))$ by exact Newton inversion of $\mathrm{ch}(E_\pm)e^{th}$ in the Schubert ring $\mathbb{Q}[h, s_2]/(h^4 - 3h^2s_2 + s_2^2,\ h^3s_2 - 2hs_2^2)$ at $t = 1, 2$ and verifies the invariance. The Schubert translation $h^2 = \sigma_2 + \sigma_{1,1}$, $s_2 = \sigma_{1,1}$ (the corpus integral receipt $\int s_2h^4 = 2 = \int \sigma_{1,1}h^4$) gives $\alpha = \frac{2}{5}\sigma_2 - \frac{3}{5}\sigma_{1,1}$. The geometric realization: the degeneracy loci of general sections of tautological bundles on the Grassmannian are Schubert-type subvarieties; the class identity is the certified Porteous computation. □

7 The weld and the FS-lock

Theorem 7.1 (The weld and the FS-lock)**.** *The state of the campaign after this round:*

(i) ***The certified scope.*** *Every rational* (p, p) *class is algebraic on: the* 114 *committed worlds (R46); the seeds' worlds (*$\mathbf{P}^2$ *and* $\mathrm{Gr}(2, 5)$*, by Theorem 6.2's derived landings); the frame's mixed sector at its committed scope (R32/R41, the AJ-correspondence equivalence). The input axis of the true input space carries the generalized fold-birth at every non-residue Albert cell (Theorems 3.1–5.1): the five fold equations, the* (p, p)*-gate, the projector spread, at every even-degree letter.*

(ii) ***The FS-lock.*** *The remaining falsifiable line for Q2 — the Clay statement at the full generality of every smooth projective variety — is the* single *law*

FS : $\quad m^*(X, p, \alpha) > 0$ *for every rational* (p, p) *class:* $\exists$ *an integrable pure pair* (E_+, E_-) *on* X *with* ch(

together with its letter-world avatar (the realization of the cohomological norm pairs of Theorem 4.1 by actual K^0*-classes). By the corpus's certified packet-equivalence (the gr25 non-circularity remark: the input is equivalent in strength to the Hodge conjecture, both directions; the R32 remark), FS* $\iff$ *Q2.*

(iii) ***The dissolution of the pair.*** $(C1, C2)$ *is dissolved:* $C2$ *is dead (Theorem 2.1) and replaced by the true stratification;* $C1$*'s conduit is a theorem at the seeds (Theorem 6.2); the one wall — the pair-realization* Σ *— is named once, as FS.*

Proof. (i) is the corpus's receipts plus this round's Theorems 6.2 and 3.1–5.1. (ii) The equivalence receipts are the gr25 dossier's non-circularity remark and the R32 remark, both certified in prior rounds: a pure pair at a rational (p,p) class exists iff the class is algebraic (the forward direction by the conduit, Lemma 6.1; the backward direction by resolving algebraic cycles into pairs). (iii) collects Theorems 2.1 and 6.2. □

8 The certificate

`albert_fold_birth_round49_checks.py`: exact arithmetic (`Fraction` only, zero floats), deterministic double-run, md5 seal, timer-stripped, with layers:

- **X0** environment: the float-scan, the import discipline.
- **X1** the death certificate: the cubic-fourfold Betti bookkeeping ($H^3(\mathbf{P}^5)=0 \Rightarrow H^3(X)=0$ by Lefschetz hyperplane; $b_4=23$ with $h^{3,1}=h^{1,3}=1$, $h^{2,2}=21$); the $A_1\times A_2$ Künneth/level decomposition; the non-embedding $\mathbb{Q}(\sqrt{D}) \not\hookrightarrow \mathbb{Q}\oplus\mathbb{Q}$ for a family of D; the countable-union meagreness model.
- **X2** the Albert layer: the five cells; the CM even-degree law; the type table (center, $\mathbb{R}$-realization, involution) per Albert type; the degree table of the R47/R48 strata.
- **X3** the orbit law: the pair identity $(\varepsilon i)(-\varepsilon i)=+1$ over all sign patterns and all even $n\le 24$ (exact $\mathbb{Z}/4$ arithmetic on i-powers); the balanced type arithmetic $(np-k,\ np-n+k)$ with $k=n/2$; the alternating trace law $\mathrm{tr}(vy\sigma x) = -\mathrm{tr}(vx\sigma y)$ over quadratic, biquadratic and quartic cyclotomic fields (exact $\mathbb{Q}$-bases); the norm rationality $N_{K/\mathbb{Q}}(x)\in\mathbb{Q}$; the projector idempotence and orbit sums over the same fields.
- **X4** the quaternion layer: the algebras (a,b) as exact 4-dimensional $\mathbb{Q}$-algebras; the pure-square law $\gamma^2 = ax^2+by^2-abz^2$; the Hamilton definiteness; the deck $\gamma^2=-1$ at $(-1,2)$; the alternating law $\mathrm{trd}(\gamma x\bar{x})=0$; the Rosati positivity $\mathrm{trd}(d\bar{d}) = 2\,\mathrm{Nrd}(d) > 0$ at the Hamilton algebra; the Hilbert symbol ramification classification over the primes ≤ 100 (exact quadratic-residue arithmetic).
- **X5** the conduit: the Newton conductor verified by exact symmetric-function arithmetic at $p\le 8$ (power sums to elementary symmetric functions, the triangular induction); the twist cancellation at the instance level.
- **X6** the seeds: the full $\mathrm{Gr}(2,5)$ computation of the exploration (the ring, the 10-row ch-table against the corpus, the ξ_{360} purity, the pair split, the three-twist Porteous difference, the Schubert landing); the $\mathbf{P}^2$ pair.
- **X7** the weld: the scope/receipt matrix and the FS-lock composition.
- **H1**–**H10** the hostile battery: sabotages (delete a stratum, flip an orbit sign, rig the Betti table, fake the conductor, corrupt the ring relations, swap the pair split, fake the twist, instantiate a fake landing, break the count seal, insert a float) — each detected by a redundant independent layer.

9 The audit protocol

AUDIT-49: a fresh-agent dispatch (the fifth fully independent audit in campaign history), armed with the adversarial-proof-audit checklist (claim match, hidden strengthenings, circular reductions, uniformity, definition drift, computational leaps, case leaks, citation accuracy, hostile counterexamples, blocked-route smell). One loop: FAIL → repair → PASS. The audit re-derives: the Lefschetz hyperplane vanishing, the Künneth level decomposition, the alternating trace identity, the quaternion arithmetic, the Newton conductor, the two-twist Porteous difference, and runs its own sabotage battery against the certificate.

10 Scope discipline

Four classes, stated plainly:

(a) *Standard mathematics, used at proven scope*: the Lefschetz hyperplane theorem; Künneth and level bookkeeping; Lefschetz $(1,1)$; the Weil operator's type action; Galois descent; Newton's identities; the splitting principle; the Thom–Porteous formula (Fulton Thm. 14.4(a)); GAGA; Serre generation; Albert's classification; the Rosati positivity; Baire's theorem.

(b) *The campaign's certified receipts*: R32 (the middle extraction, the input-axis decomposition); the censuses (R35–R41, R43); R44/R45; R46 (the leaf discharge at all 114 worlds); R47 (the RM fold-birth); R48 (the UC fold-birth, the sign-blind law); the gr25 dossier (Theorem C, the ξ_{360} lattice, the minimal scale 360); the CI-green capstone (the fold-pair calculus, the principality `scaleSet_principal`, `p2_mstar_eq_two`); the packet-equivalence receipts.

(c) *This round's new mathematics*: the death certificate (Theorem 2.1); the Albert stratification transfer (Theorem 3.1); the orbit law (Theorem 4.1); the quaternion deck (Theorem 5.1); the conduit's assembled core (Lemma 6.1); the derived landing cycles (Theorem 6.2); the FS-lock (Theorem 7.1).

(d) *The commitment class*: FS — the pair-realization, one named law, equivalent in strength to the Clay statement by the corpus's own receipts; the letter-world realization step (the norm pairs of Theorem 4.1 as K^0-classes) is the same wall photographed from the letter side. No receipt is silently strengthened; the closing string is emitted at exactly the scope of (i)–(iii).

The quantifier line: **the false hypothesis is dead; the true input space is stratified; the fold equations hold at every even-degree letter in the true stratification; the landing cycles are derived at both seeds including the open-sector grade; the remaining falsifiable line for Q2 is the single law FS.** The campaign continues per the operator's standing order.